

ALI BASELIZADEH

Interdisciplinary Researcher — Computational Neuroscience, Computational Psychiatry & Neurophilosophy

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RESEARCH PROFILE

Interdisciplinary researcher bridging electrical engineering, philosophy of science, and human-centered AI (current MSc, University of Milan), working toward mathematically grounded, computational models of psychiatric and neurological disorders. Research interests center on computational psychiatry, interoception, and personalized closed-loop neuromodulation (TMS/tDCS), approached through Bayesian and active inference frameworks. Applies a background in signal processing and dynamical systems (B.Sc. Electrical Engineering) to EEG analysis (MNE-Python) and Bayesian/HGF modeling. Currently leading or co-leading four research projects: a tDCS meta-analysis, a TMS-EEG perturbational complexity (PCI) scoping review, a resting-state EEG complexity study, and a computational active inference account of interoception in Zar healing rituals. Co-author of an Alzheimer's biomarker manuscript under review at BMC Geriatrics (Springer Nature). Seeking a fully funded PhD position in computational neuroscience, computational psychiatry, or neurophilosophy, with a target start around Fall 2027 or the following cycle. In the interim, also open to Research Assistant or laboratory technician roles that would allow hands-on contribution to ongoing research while preparing for doctoral study.

EDUCATION

M.Sc., Human-Centered Artificial Intelligence — Sept 2025 – Present

University of Milan, Milan, Italy

M.Sc., Philosophy of Science — Sept 2016

Amir Kabir University of Technology, Tehran, Iran

B.Sc., Electrical Engineering (Telecommunications) — Sept 2013

K.N. Toosi University of Technology, Tehran, Iran

SELECTED RESEARCH PROJECTS (IN PROGRESS)

Circuit versus Diagnosis: A Systematic Review and Meta-Regression of Cognitive Outcomes following Transcranial Direct Current Stimulation (tDCS) Across Psychiatric Disorders

Co-investigator; PRISMA protocol testing whether stimulation site (circuit) outperforms diagnostic category in predicting tDCS effects on working memory, inhibitory control, and cognitive flexibility.

Mapping TMS-EEG Perturbational Complexity (PCI) Across Disorders of Consciousness, Schizophrenia, Ego Dissolution, and Alzheimer's Disease: A Scoping Review

Systematically maps PCI and related complexity indices as consciousness markers across these conditions.

Resting-State EEG Complexity as a Cross-Population Biomarker: An Analysis of Two Open Datasets

Tests generalizability of complexity-based EEG biomarkers across the ASZED and Olejarczyk & Jernajczyk open datasets.

Respiratory Interoception and Precision Re-weighting in Zar Healing Rituals: A Computational Active Inference Account

With Prof. Maryam Gharasou (ethnomusicology, University of Tehran); frames respiratory interoception and precision re-weighting as the mechanism behind Zar/Ahl-e Hava trance healing, with possession belief as a cultural prior.

WORKING PAPERS

“Modeling the Fragmented Self: Sense of Agency and Ego Dissolution in Psychiatric Contexts”

Philosophy essay grounded in Haggard's comparator model of motor cognition, examining disruptions of the sense of agency in psychiatric disorders and psychedelic-induced ego dissolution. Writing sample available on request.

PUBLICATIONS

Shahkarami, F., Nasiri, H., Najafi, Y., Changizi, S., Khosravi, F., **Baselizadeh, A.**, Shakeri, S., Akhgari, A., Kermanizadeh Tavakoli, M., Mahmoudvand, B., Gholibeigian, Z., Hashemi, M., Jafari Razi, Z., Mayeli, M. “Elevated CSF GAP-43 Linked to Cognitive Impairment Through Increased Anatomical Temporal Lobe Atrophy in Mild Cognitive Impairment.” *BMC Geriatrics* (Springer Nature) — Under Review.

Contribution: literature review and drafting of manuscript sections.

RESEARCH EXPERIENCE

Research Assistant — Neuropsychology Association — 2020–2023; 2025–Present

Students' Scientific Research Center, Tehran University of Medical Sciences

- Conducted research on Alzheimer's disease using large-scale neuroimaging and clinical datasets (ADNI), resulting in a peer-reviewed submission.
- Co-authoring and mentorship of an interdisciplinary book on brain health and disease, covering aging, neurodegeneration, mood, and psychotic disorders.

Research Assistant — Interdisciplinary Schools — 2021–2023

Sharif University of Technology

- Contributed to research on visual-motor learning, including a systematic review and an original study.
- Performed scientific data collection and curation using Scopus, Web of Science, and PubMed.

COMPUTATIONAL & TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, MNE-Python for EEG; Bayesian/HGF modeling for computational-psychiatry analyses), MATLAB (signal & image processing)

Data & statistics: SPSS, Power BI, SQL (queries, joins)

Engineering foundation: signal processing, dynamical systems, probability and statistics (B.Sc. Electrical Engineering)

General: Microsoft Office 365 (Excel, PowerPoint, Word)

TEACHING EXPERIENCE

- Teaching Assistant, Philosophy of Mind, Philosophy of Science — Amir Kabir University of Technology (2015)
- Teaching Assistant, Differential Equations, Electrical Engineering — K.N. Toosi University of Technology (2011)
- Teaching Assistant, Engineering Statistics and Probability, Electrical Engineering — K.N. Toosi University of Technology (2010)

HONORS & AWARDS

- Ranked 557th among 311,000+ participants (top 1%) — Iran National Universities Entrance Exam
- Ranked 14th among 500+ participants — Iran National M.Sc. Entrance Exam
- Top 1% of 220,256 participants — Iran National Entrance Exam

SELECTED CERTIFICATIONS

- Computational Psychiatry Course (CPC) — Translational Neuromodeling Unit, University of Zurich/ETH Zurich (enrolled, online; 1–7 Sept 2026)
- Computational Neuroscience — University of Washington (2023)
- QEEG: From Theory to Application — Tehran University of Medical Sciences (Nov 2020)
- Python Programming in Neuroscience and Cognitive Science — Iran University of Medical Sciences (Aug 2021)
- Diffusion Tensor Imaging Workshop — Tehran University of Medical Sciences (2021)

Additional relevant coursework: Thinking AI: Ethical, Legal and Social Aspects of AI (Sorbonne Université, 2026); Neuroscience and Neuroimaging Specialization (Johns Hopkins University, 2023); Philosophy and the Sciences: Introduction to the Philosophy of Cognitive Sciences (University of Edinburgh, 2022). Complete list available upon request.

LANGUAGE SKILLS

- English — Professional Working Proficiency
- Persian — Native Proficiency
- German — Limited Working Proficiency
- Italian — Limited Working Proficiency
- Arabic — Limited Working Proficiency

MEMBERSHIPS

Iran National Brain Mapping Centre (2022–Present)

REFERENCES

- Maryam Gharasou — Associate Professor, College of Fine Arts, Faculty of Performing Arts and Music, University of Tehran. maryam.gharasou@ut.ac.ir
- Mahsa Mayeli, MD, MPH — Postdoctoral Associate, Department of Radiology & Biomedical Imaging, School of Medicine, Yale University. mahsa.mayeli@yale.edu